CHOW'S LEMMA FOR ALGEBRAIC SPACES: A BLOWUP OF A SEPARATED FINITE TYPE SPACE EMBEDS INTO PROJECTIVE SPACE OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9274-9290 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 9292-9365, the complete author proof (72 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The chapter cites Knutson, IV Theorem 3.1 in a reference block; that citation is part of the statement block and no bibliography entry is required for the collected spans (no \cite outside the reference environment).

Editorial note. The statement's reference environment cites \cite[IV Theorem 3.1]{Kn}; a bibliography.tex holding the snapshot my.bib Kn entry (Knutson, Algebraic spaces, LNM 203, Springer-Verlag 1971) is attached (precedent S3-morph-etale-nbhd-dominates-smooth EGA4).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Approximation and descent of spaces and morphisms of finite presentation (Limits of Spaces, proposition-approximate, lemmas descend-separated, descend-finite-presentation, descend-separated-morphism, topology-limit), the schemes-level Chow input and the H-projectivity criteria from the Morphisms chapter (projective-over-quasi-projective-is-H-projective, H-projective-composition); cited inputs with the reduction and diagram construction written in the retained span.

Source reference. Original lemma 088S: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9225–9244. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01WE: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9365–9365. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 087S: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9365–9365. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07SK: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9365–9365. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 084T: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9365–9365. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0851: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9365–9365. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 086V: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9365–9365. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07SU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9365–9365. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 082Z: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9365–9365. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1 (Chow's lemma)**.** [1, IV Theorem 3.1] *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f separated of finite type, and Y separated and Noetherian. Then there exists a commutative diagram*

$$\begin{array}{ccccc} X & \longleftarrow & X' & \longrightarrow & \mathbf{P}^n_Y \\ & \searrow & \downarrow & \swarrow & \\ & & Y & & \end{array}$$

where $X' \to X$ is a U-admissible blowup for some dense open $U \subset X$ and the morphism $X' \to \mathbf{P}^n_Y$ is an immersion.

Proof. In this first paragraph of the proof we reduce the lemma to the case where Y is of finite type over $\mathrm{Spec}(\mathbf{Z})$. We may and do replace the base scheme S by $\mathrm{Spec}(\mathbf{Z})$. We can write $Y = \lim Y_i$ as a directed limit of separated algebraic spaces of finite type over $\mathrm{Spec}(\mathbf{Z})$, see Limits of Spaces, Proposition 07SU and Lemma 084T. For all i sufficiently large we can find a separated finite type morphism $X_i \to Y_i$ such that $X = Y \times_{Y_i} X_i$, see Limits of Spaces, Lemmas 07SK and 0851. Let $\eta_1, \ldots, \eta_n$ be the generic points of the irreducible components of $|X|$ (X is Noetherian as a finite type separated algebraic space over the Noetherian algebraic space Y and therefore $|X|$ is a Noetherian topological space). By Limits of Spaces, Lemma 086V we find that the images of $\eta_1, \ldots, \eta_n$ in $|X_i|$ are distinct for i large enough. We may replace X_i by the scheme theoretic image of the (quasi-compact, in fact affine) morphism $X \to X_i$. After this replacement we see that the images of $\eta_1, \ldots, \eta_n$ in $|X_i|$ are the generic points of the irreducible components of $|X_i|$,

see Morphisms of Spaces, Lemma 082Z. Having said this, suppose we can find a diagram

$$\begin{array}{ccccc} X_i & \longleftarrow & X_i' & \longrightarrow & \mathbf{P}^n_{Y_i} \\ & \searrow & \downarrow & \swarrow & \\ & & Y & & \end{array}$$

where $X_i' \to X_i$ is a U_i-admissible blowup for some dense open $U_i \subset X_i$ and the morphism $X_i' \to \mathbf{P}^n_{Y_i}$ is an immersion. Then the strict transform $X' \to X$ of X relative to $X_i' \to X_i$ is a U-admissible blowing up where $U \subset X$ is the inverse image of U_i in X. Because of our carefully chosen index i it follows that $\eta_1, \ldots, \eta_n \in |U|$ and $U \subset X$ is dense. Moreover, $X' \to \mathbf{P}^n_Y$ is an immersion as X' is closed in $X_i' \times_{X_i} X = X_i' \times_{Y_i} Y$ which comes with an immersion into $\mathbf{P}^n_Y$. Thus we have reduced to the situation of the following paragraph.

Assume that Y is separated of finite type over $\mathrm{Spec}(\mathbf{Z})$. Then $X \to \mathrm{Spec}(\mathbf{Z})$ is separated of finite type as well. We apply Lemma 088S to $X \to \mathrm{Spec}(\mathbf{Z})$ to find a dense open subspace $U \subset X$ and a commutative diagram

$$\begin{array}{ccccccc} & & & & U & & \\ & \swarrow & & \downarrow & & \searrow & \\ X & \longleftarrow & X' & & \longrightarrow & Z' & \longrightarrow & Z \\ & \searrow & & \downarrow & & \swarrow & & \searrow \\ & & & \mathrm{Spec}(\mathbf{Z}) & \longleftarrow & \mathbf{P}^n_{\mathbf{Z}} & & \end{array}$$

with all the properties listed in the lemma. Note that Z has an ample invertible sheaf, namely $\mathcal{O}_{\mathbf{P}^n}(1)|_Z$. Hence $Z' \to Z$ is a H-projective morphism by Morphisms, Lemma 087S. It follows that $Z' \to \mathrm{Spec}(\mathbf{Z})$ is H-projective by Morphisms, Lemma 01WE. Thus there exists a closed immersion $Z' \to \mathbf{P}^m_{\mathrm{Spec}(\mathbf{Z})}$ for some $m \geq 0$. It follows that the diagonal morphism

$$X' \to Y \times \mathbf{P}^m_{\mathbf{Z}} = \mathbf{P}^m_Y$$

is an immersion (because the composition with the projection to $\mathbf{P}^m_{\mathbf{Z}}$ is an immersion) and we win. □

References

[1] Knutson, Donald. *Algebraic spaces*. Volume 203 of Lecture Notes in Mathematics. Springer-Verlag, 1971.